

Routing Information Protocol:

Routing Information Protocol (RIP) is a dynamic routing protocol used in local and wide area networks. Here are the key features and concepts associated with RIP:

Key Features

- 1. Distance-Vector Routing Protocol**
- 2. Hop Limit**
- 3. Periodic Updates**
- 4. Simple Metric**
- 5. Split Horizon**
- 6. Hold-Down Timers**

Distance-Vector Routing Protocol: RIP uses the distance-vector algorithm to determine the best path to a destination network. It measures the distance in terms of the number of hops.

Hop Limit: The maximum number of hops allowed for RIP is 15.

Periodic Updates: RIP routers send updates to their neighboring routers every 30 seconds. These updates contain the entire routing table.

Simple Metric: RIP uses a simple hop count as its metric. This can lead to suboptimal routing if the shortest path in terms of hops is not the fastest path.

Split Horizon: Split horizon is a method used by distance vector protocols to prevent network routing loops. Never send routing information back in the direction from which it was received.

Versions of RIP

RIP Version 1 (RIPv1):

- Classful protocol: Does not support subnetting, which means it doesn't send subnet mask information with its routing updates.
- Broadcasts updates to all hosts.

RIP Version 2 (RIPv2):

- Classless protocol: Supports subnet masks and VLSM (Variable Length Subnet Mask).
- Uses multicast for sending updates to reduce unnecessary load on all hosts.
- Includes authentication for updates, which adds a layer of security.

RIP:

- RIP is a Distance Vector Routing Protocol
- **AD Value: 120**
- **Best Path Selection Mechanism:** Least Hop Count
- If Multiple path with Same Bandwidth and same hop count, then best path selected randomly or the one which advertise first
- **Algorithm Used:** Bellmen Ford
- **Max Hop Count: 15**
- By Default RIP Version 1 has been enabled
- **Loop Prevention Mechanism** - poison reverse
- **Command used to check details about the Protocols:**

Router# Show ip Protocols

Versions of RIP

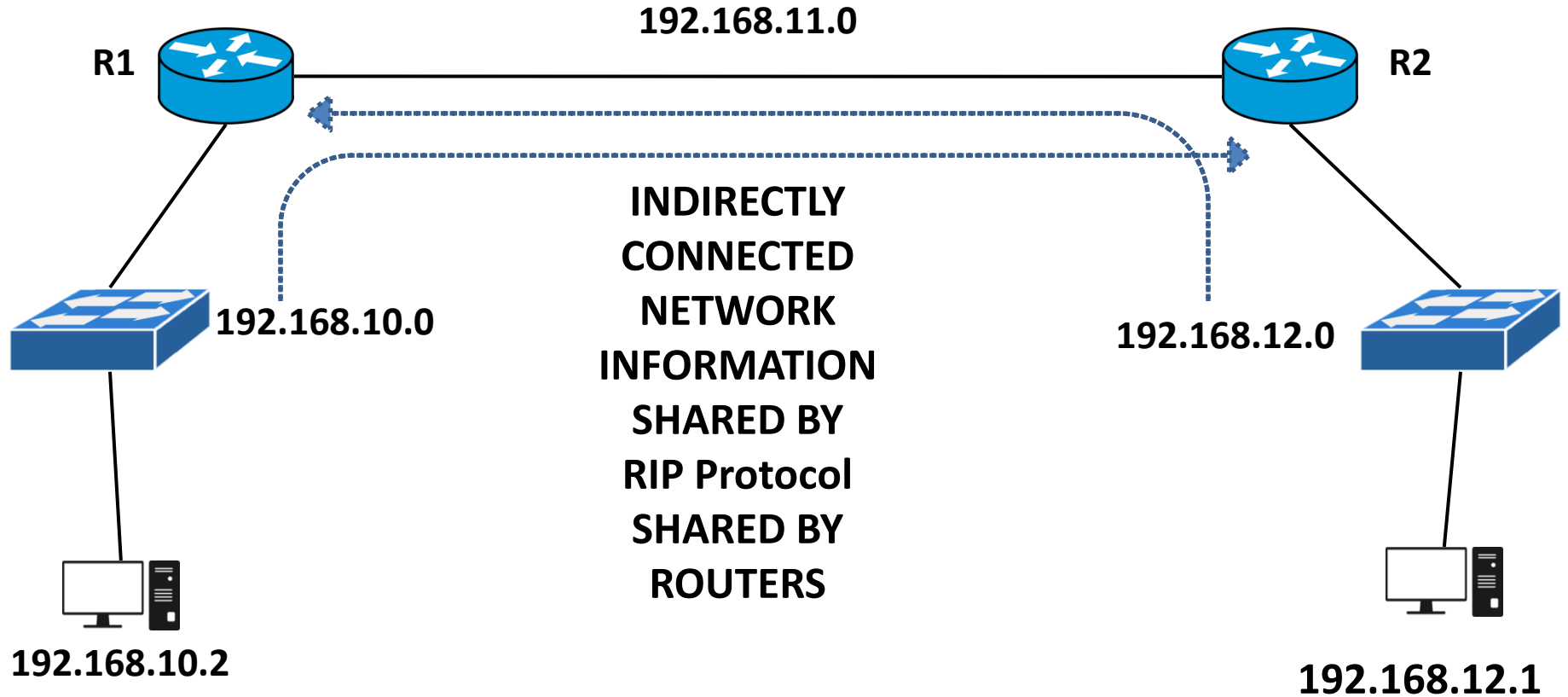
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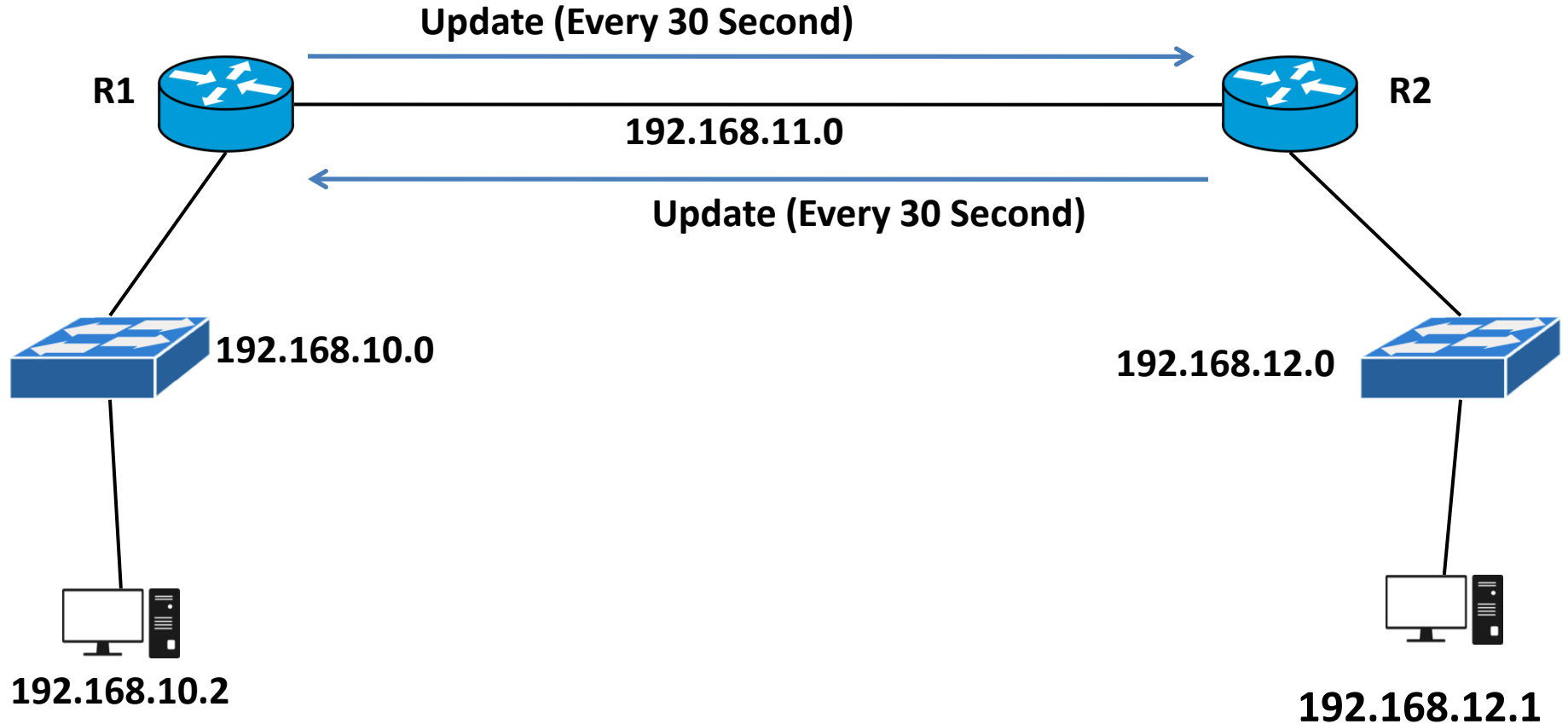
RIP Version 2 (RIPv2):

- Classless protocol: Supports subnet masks and VLSM (Variable Length Subnet Mask).
- Uses multicast for sending updates to reduce unnecessary load on all hosts.
- Includes authentication for updates, which adds a layer of security.

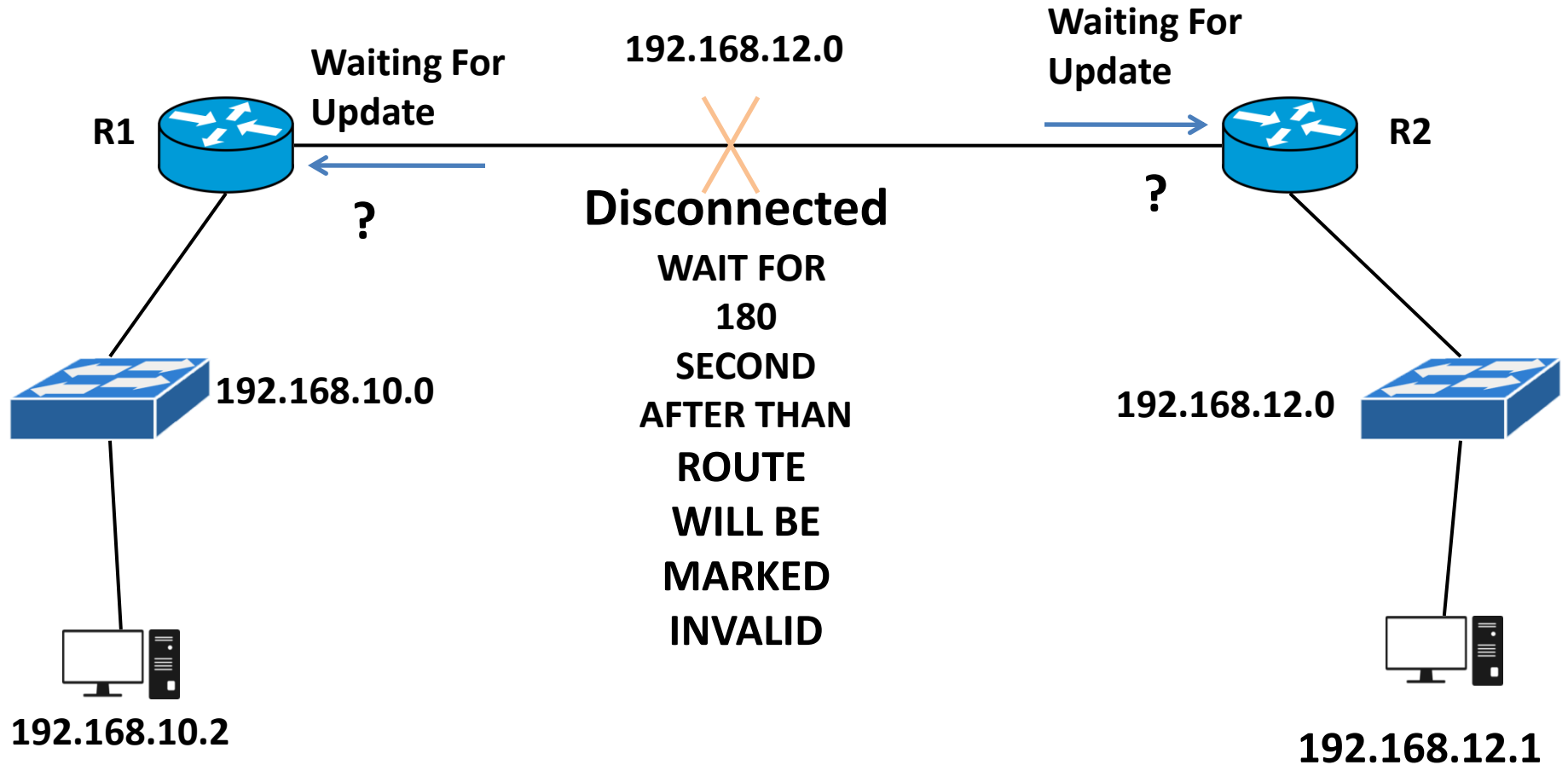
Dynamic Routing:



RIP Update Timer 30 Second:



RIP Invalid Timer 180 Second:



Invalid Timer in RIP:

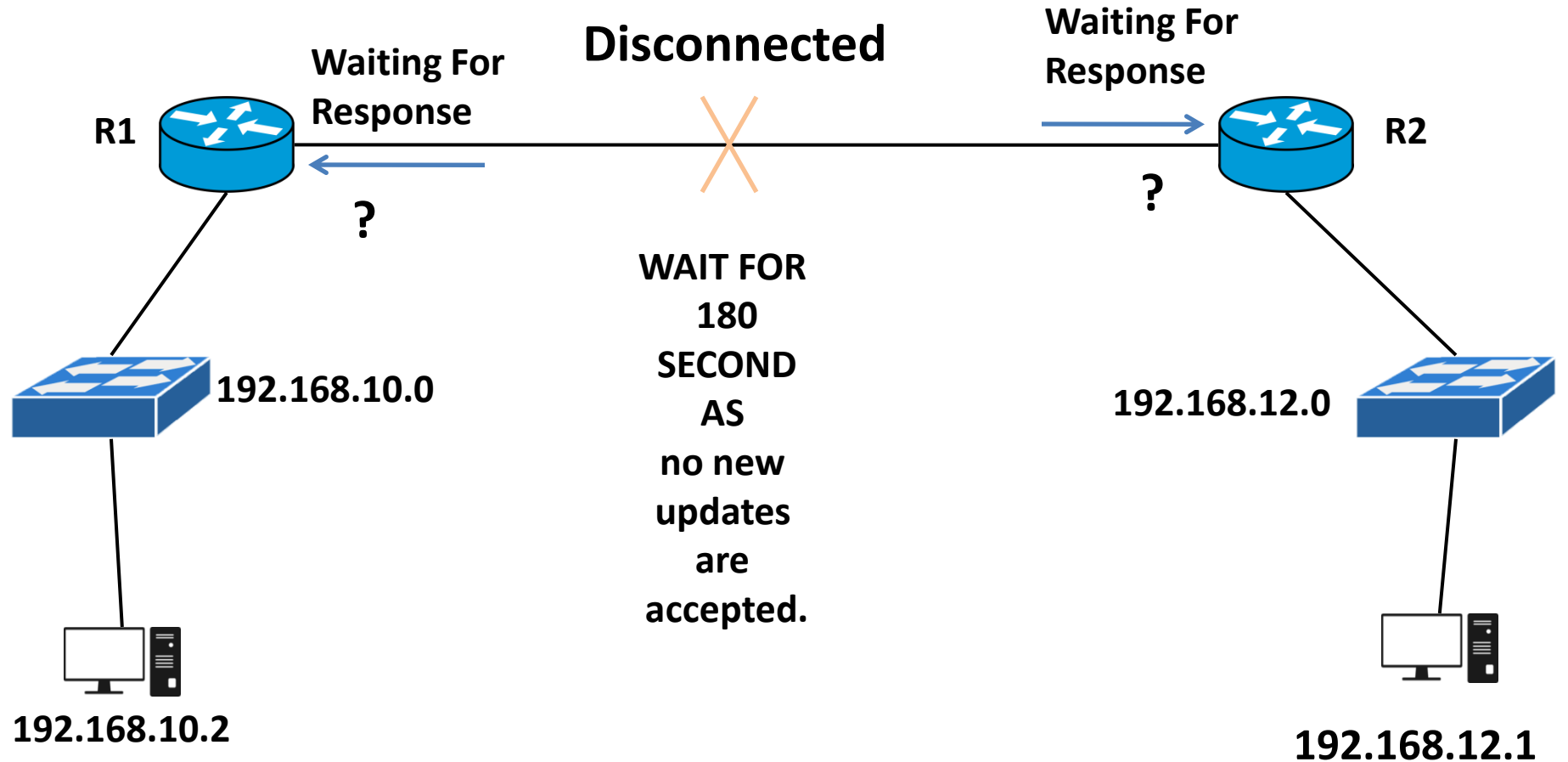
Purpose:

The Invalid Timer determines how long a route can stay in the routing table without receiving an update before it is marked as invalid.

Functionality:

- When a route is first learned or updated, the Invalid Timer is set.
- If no updates are received for that route before the Invalid Timer expires, the route is considered invalid.
- The route is not immediately removed from the routing table but is marked as unreachable. <https://www.youtube.com/@techwithravish>

RIP Hold On Timer 180 Second:



Hold-down Timer

Purpose:

The Hold-down Timer is used to stabilize the network by preventing the router from accepting any new, possibly incorrect, routing information about an invalid route for a specific period.

Functionality:

- Once a route is marked as invalid (due to the Invalid Timer expiring), the Hold-down Timer is activated.
- During the Hold-down period, the router will ignore any updates for that route, even if they come from different neighbors.
- This helps to avoid routing loops and flapping, where routes might repeatedly switch between valid and invalid states.

Basic Configuration of RIPv1

Enable RIP Routing Process:

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router rip
```

Specify Networks:

```
Router(config-router)# network 192.168.10.0
```

```
Router(config-router)# network 192.168.11.0
```

```
Router(config-router)# exit
```

Verify Configuration:

```
Router# show ip protocols
```

```
Router# show ip route
```

Basic Configuration of RIPv2

Enable RIP Routing Process:

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router rip
```

Enable Version 2:

```
Router(config-router)# version 2
```

Specify Networks:

```
Router(config-router)# network 192.168.10.0
```

```
Router(config-router)# network 192.168.11.0
```

```
Router(config-router)# exit
```